
title: “rnaseq-toolkit: an R/Python package for streamlined differential expression analysis and pathway enrichment visualization from RNA-seq data” authors:

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Abstract

Motivation: RNA sequencing (RNA-seq) has become a standard technique for transcriptomic profiling. However, the analysis pipeline—from raw count matrices to differential expression analysis (DEA) and pathway enrichment—remains fragmented across different programming languages and tools, requiring significant computational expertise. **Results:** We present `rnaseq-toolkit`, an open-source Python package that provides a unified, switchable interface for RNA-seq analysis. It integrates state-of-the-art methods for normalization, DEA (DESeq2, edgeR-like), and pathway enrichment (GSEA, GO, KEGG) into a single, reproducible pipeline. `rnaseq-toolkit` automatically generates publication-quality visualizations and self-contained HTML reports, significantly reducing the time from data acquisition to biological insight. We demonstrate its utility and performance using public datasets, including COVID-19 transcriptomic profiles. **Availability and implementation:** `rnaseq-toolkit` is implemented in Python 3.9+ and is freely available under the MIT license at <https://github.com/rnaseq-toolkit/rnaseq-toolkit>. A Docker image and Snakemake workflow are provided for seamless integration into existing bioinformatics pipelines.

1. Introduction

Transcriptomic analysis via RNA-seq is essential for understanding cellular responses in health and disease [1]. The standard analytical workflow typically involves read alignment, quantification, normalization, differential expression analysis (DEA), and functional

enrichment [2]. While excellent individual tools exist—such as DESeq2 [3] and edgeR [4] for DEA, and clusterProfiler [5] or gseapy [6] for enrichment—they often require complex custom scripts to integrate. This fragmentation poses a barrier to entry for many biologists and reduces the reproducibility of bioinformatics analyses.

To address these challenges, we developed `rnaseq-toolkit`, a comprehensive Python package that unifies the RNA-seq analysis pipeline. By providing a standardized interface to multiple normalization and DEA methods, it allows researchers to easily compare analytical strategies. Furthermore, the automated generation of interactive HTML reports and publication-ready visualizations accelerates the interpretation of results.

2. Features and Implementation

`rnaseq-toolkit` is designed around a central `RNAseqPipeline` class that orchestrates the entire workflow. It accepts raw count matrices and sample metadata as input, and executes the following steps:

- 1. Data Preprocessing and Normalization:** The toolkit filters low-expressed genes based on user-defined thresholds. It supports multiple normalization strategies, including median-of-ratios (default for DESeq2), Trimmed Mean of M-values (TMM, standard for edgeR), Counts Per Million (CPM), and Variance Stabilizing Transformation (VST). Users can switch between these methods using a single argument (`--norm-method`).
- 2. Differential Expression Analysis:** The core DEA is powered by PyDESeq2 [7], a robust Python implementation of the DESeq2 algorithm. Alternatively, users can select an edgeR-like quasi-likelihood approach implemented via `statsmodels` [8] Negative Binomial Generalized Linear Models (GLM). Both methods return results in a standardized format, facilitating direct comparison.
- 3. Pathway Enrichment:** Significant differentially expressed genes (DEGs) are automatically analyzed for functional enrichment. `rnaseq-toolkit` performs pre-ranked Gene Set Enrichment Analysis (GSEA) [9] and over-representation analysis for Gene Ontology (GO) and KEGG pathways using `gseapy` [6], which interfaces with the Enrichr database [10].
- 4. Visualization and Reporting:** The package generates high-quality (300 DPI) plots, including volcano plots, MA plots, Principal Component Analysis (PCA), clustered

heatmaps of top DEGs, and GSEA dotplots. These visualizations, along with interactive data tables, are compiled into a self-contained HTML report.

The toolkit can be executed via a command-line interface (CLI), directly within Python scripts, or through a provided Snakemake [11] workflow for scalable execution on high-performance computing clusters. A Docker container is also available to ensure complete computational reproducibility.

3. Application and Performance

To validate `rnaseq-toolkit`, we analyzed three distinct bulk RNA-seq datasets: a synthetic dataset, a COVID-19 patient cohort (GEO accession GSE157103) [12], and a Neuronal Ceroid Lipofuscinosis (Batten disease) mouse model (GSE123509) [13]. Using the CLI, the entire pipeline—from raw counts to the final HTML report—can be executed with a single command:

```
rnaseq-toolkit \
  --counts GSE123509_counts.csv \
  --metadata GSE123509_metadata.csv \
  --design "~condition" \
  --contrast condition Batten control \
  --norm-method deseq2 \
  --dea-method deseq2 \
  --output results/
```

The analysis of the Batten disease model (Tpp1-/- vs. wildtype) successfully identified key neurodegenerative markers. Notably, `Foxg1` was significantly upregulated ($\log_2 \text{FC} = 10.87$, $p_{adj} < 10^{-24}$), while Purkinje cell markers `Gabra6` ($\log_2 \text{FC} = -10.66$) and `Pcp2` ($\log_2 \text{FC} = -10.59$) were strongly downregulated, consistent with the severe cerebellar degeneration characteristic of NCL. Furthermore, automated Gene Set Enrichment Analysis (GSEA) on the Batten disease dataset highlighted significant alterations in the “Neuroactive ligand-receptor interaction” pathway (Normalized Enrichment Score = 1.74, FDR < 0.05), providing immediate biological context to the gene-level findings. Similar enrichment analysis on the COVID-19 dataset correctly identified “Phosphatidylinositol signaling system” and immune-related pathways among the top enriched terms.

We performed a comprehensive benchmark comparing the PyDESeq2 implementation within `rnaseq-toolkit` against the edgeR-like method across all three datasets (Table 1, Figure 1). The results demonstrated high concordance in log2 fold-change estimates for the synthetic ($r = 1.00$) and COVID-19 datasets ($r = 0.89$). Interestingly, in the Batten disease dataset (n=3 per group), the concordance was lower ($r = 0.396$), highlighting the sensitivity of different statistical models to small sample sizes and emphasizing the value of `rnaseq-toolkit`'s ability to easily switch between DEA methods for robust validation.

Dataset	Method	DEGs (padj<0.05)	Runtime (s)	LFC Pearson r	Jaccard Index
Synthetic	DESeq2	104	7.8	1.000	0.915
Synthetic	edgeR-like	99	8.1	1.000	0.915
COVID-19	DESeq2	113	60.2	0.890	0.154
COVID-19	edgeR-like	217	90.5	0.890	0.154
Batten/NCL	DESeq2	4829	55.5	0.396	0.662
Batten/NCL	edgeR-like	4252	181.0	0.396	0.662

Table 1: Benchmark of DESeq2 vs edgeR-like DEA methods across three independent datasets.

Figure 1: Benchmark Summary

Figure 1: Comparison of the number of DEGs, runtime, and method concordance (LFC Pearson correlation and Jaccard index).

4. Conclusion

`rnaseq-toolkit` streamlines the RNA-seq analysis workflow, making robust statistical methods and advanced visualizations accessible through a unified interface. By automating routine tasks and generating comprehensive reports, it enables researchers to focus on biological interpretation rather than software engineering. Future development will focus on integrating single-cell RNA-seq (scRNA-seq) capabilities and expanding the suite of supported enrichment databases.

References

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